2.6. Algebras defined by generators and relations

If $f_1, \dots, f_m$ are elements of the free algebra $k\langle x_1, \dots, x_n\rangle$, we say that the algebra $A := k\langle x_1, \dots, x_n\rangle/\langle\{f_1, \dots, f_m\}\rangle$ is **generated** by $x_1, \dots, x_n$ with **defining relations** $f_1 = 0, \ \dots, \ f_m = 0$.

2.7. Examples of algebras

The following two examples are among the simplest interesting examples of noncommutative associative algebras:

(1) the **Weyl algebra**, $k\langle x, y\rangle/\langle yx - xy - 1\rangle$;

(2) the q**-Weyl algebra**, generated by x, x^{-1}, y, y^{-1} with defining relations $yx = qxy$ and $xx^{-1} = x^{-1}x = yy^{-1} = y^{-1}y = 1$.

Proposition 2.7.1. *(i) A basis for the Weyl algebra A is $\{x^i y^j, i, j \geq 0\}$.*

(ii) A basis for the q-Weyl algebra A_q is $\{x^i y^j, i, j \in \mathbb{Z}\}$.

Proof. (i) First let us show that the elements $x^i y^j$ are a spanning set for A. To do this, note that any word in x, y can be ordered to have all the x's on the left of the y's, at the cost of interchanging some x and y. Since $yx - xy = 1$, this will lead to error terms, but these terms will be sums of monomials that have a smaller number of letters x, y than the original word. Therefore, continuing this process, we can order everything and represent any word as a linear combination of $x^i y^j$.

The proof that $x^i y^j$ are linearly independent is based on representation theory. Namely, let a be a variable, and let $E = t^a k[a][t, t^{-1}]$ (here t^a is just a formal symbol, so really $E = k[a][t, t^{-1}]$). Then E is a representation of A with action given by $xf = tf$ and $yf = \frac{df}{dt}$ (where $\frac{d(t^{a+n})}{dt} := (a+n)t^{a+n-1}$). Suppose now that we have a nontrivial linear relation $\sum c_{ij} x^i y^j = 0$. Then the operator

$$L = \sum c_{ij} t^i \left(\frac{d}{dt}\right)^j$$